

4. AUTOMATIC CONTROLLERS

Industrial controllers may be classified according to their control actions as

- i) On-off controllers
- ii) Proportional controllers
- iii) Integral controllers
- iv) Proportional-plus-Integral controllers
- v) Proportional-plus-Differential controllers
- vi) Proportional-plus-Integral-plus-Differential Controllers

Most of the industrial controllers use electricity or pressurised fluids like oil or air as power sources. Accordingly, controllers may also be classified as electronic controllers, hydraulic controllers or pneumatic controllers. The choice of the controller is based on the nature of plant and the operating conditions, safety, cost, reliability, availability, accuracy, weight, size

On-off controllers

In on-off control (two position control) system, the actuating element has only two fixed positions, commonly on and off. Let the actuating error signal be $e(t)$ and the output signal from the controller be $u(t)$.

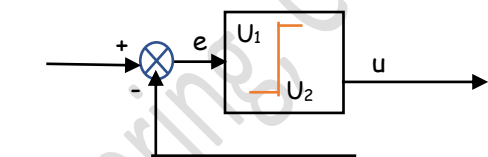


Fig. 4.1

In two position control, the signal $u(t)$ remains at either maximum or minimum value, depending on whether the actuating error signal is positive or negative.

$$u(t) = U_1, \text{ for } e(t) > 0$$

$$= U_2, \text{ for } e(t) < 0$$

Where U_1 and U_2 are constants. The minimum value U_2 is either zero or $-U_1$. On-off controllers are generally electrical devices, and an electric solenoid operated valve is widely used in such controllers.

Proportional controllers

For a controller with proportional control action, the relationship between the output of the controller $u(t)$ and the actuating error signal $e(t)$ is

$$u(t) = K_p e(t) \text{ or}$$

$$\frac{U(s)}{E(s)} = K_p \text{ where } K_p \text{ is called proportional gain. (Please note that this is different from the position error constant } K_p)$$

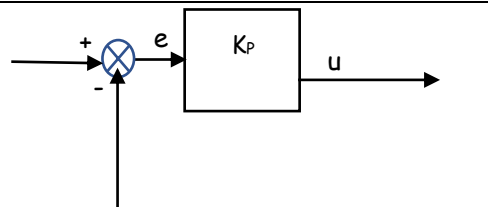


Fig. 4.2

Whatever be the actual mechanism may be or whatever the form of operating power may be, the proportional controller is essentially an amplifier with adjustable gain.

Integral controllers

In a controller with integral control action, the value of the controller output $u(t)$ is changed at a rate proportional to the actuating error signal $e(t)$

$$\frac{d}{dt} u(t) = K_i e(t)$$

$$u(t) = K_i \int e(t) dt$$

$$\text{The transfer function of the integral controller } \frac{U(s)}{E(s)} = \frac{K_i}{s}$$

Where K_i is an adjustable constant

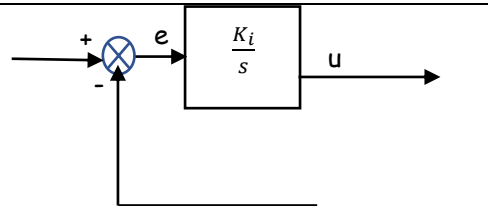


Fig. 4.3

Proportional-plus-Integral (PI)controller

The control action of a PI controller is defined by

$$u(t) = K_p e(t) + K_i \int e(t) dt$$

$$\frac{U(s)}{E(s)} = K_p + \frac{K_i}{s} = K_p \left(1 + \frac{1}{T_i s} \right)$$

Where $\frac{K_p}{T_i} = K_i$ and T_i is known as integral time

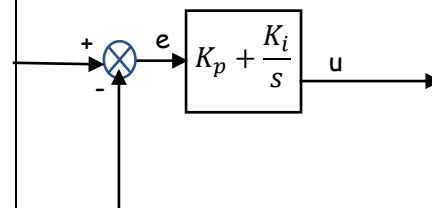


Fig. 4.4

Proportional-plus-derivative (PD) controller

The control action of a PD controller is defined by

$$u(t) = K_p e(t) + K_d \frac{d}{dt} e(t)$$

$$\frac{U(s)}{E(s)} = K_p + K_d s = K_p (1 + T_d s)$$

Where $K_p T_d = K_d$ and T_d is called derivative time

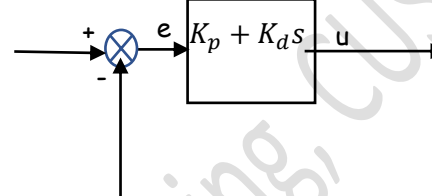


Fig. 4.5

Proportional-plus-integral-plus-derivative (PID) control

PID control action is a combination of proportional control action, integral control action, and derivative control action

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{d}{dt} e(t)$$

$$\frac{U(s)}{E(s)} = K_p + \frac{K_i}{s} + K_d s = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

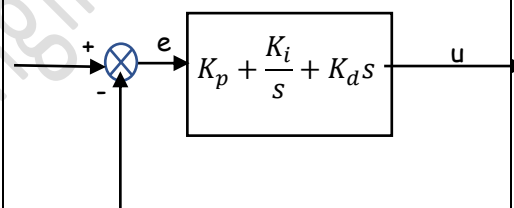


Fig. 4.6

We can get these control actions electronically, pneumatically, or hydraulically. Let us discuss how we can get these control actions electronically using operational amplifiers (op amps). You are requested to refer the text book *Modern Control Systems* by K Ogata for pneumatic and hydraulic control actions.

Operational Amplifiers (op amps) are used to amplify signals in sensor circuits. They are also frequently used in filters used for compensation purposes.

The figure shows an op amp. Usually the ground is chosen as 0 V and the input voltages e_1 and e_2 are measured relative to the ground. The input voltage e_1 is inverted at the minus terminal and e_2 is not inverted. The total input to the amplifier is $e_2 - e_1$. Therefore, the output voltage

$e_o = K(e_2 - e_1)$, where K is the differential gain (voltage gain).

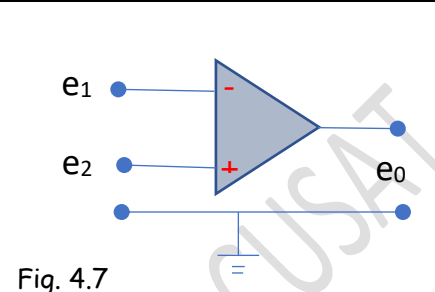


Fig. 4.7

The magnitude of K is approximately of the order of 10^5 for dc signals and low frequency (less than 10 Hz) ac signals. The value of K decreases with increase in frequency and it is approximately equal to 1 for frequencies above 1 MHz. Since op amps amplify the difference in voltages, it is also called **differential amplifier**. Since the gain of the op amp is very high, it is necessary to have negative feedback from the output to input to make the amplifier stable. The feedback is made from the output to the negative input so that the feedback becomes negative.

In an ideal op amp, no current flows in to the input terminals and the output voltage is not affected by the load connected to the output terminal. In other words, the input impedance of op amp is infinity and the output impedance is zero. In actual case, this may not be exactly true.

Inverting amplifier

Let us consider an op amp circuit as shown. The positive terminal is connected to the ground. Let e' be the potential at the branching point for i_2 .

Then,

$$i_1 = \frac{e_i - e'}{R_1}, \quad i_2 = \frac{e' - e_o}{R_2}$$

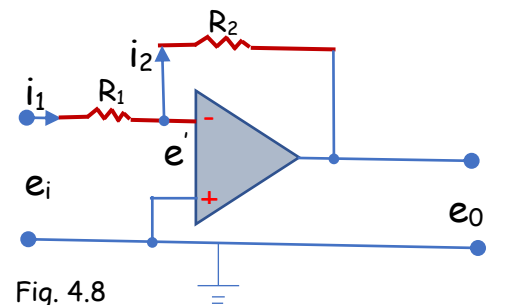


Fig. 4.8

Since for an ideal op amp, negligible current flows in to the amplifier, $i_1 = i_2$

$$\frac{e_i - e'}{R_1} = \frac{e' - e_o}{R_2} \dots \dots \dots (4.1)$$

For op amps, $e_o = K(e_2 - e_1) = K(0 - e') = -Ke'$ (Refer fig.4.7)

Since K is usually very large, e' should be very small (close to zero) so that their product gives a reasonable value for e_o . Hence equation (4.1) becomes

$$\frac{e_i}{R_1} = \frac{-e_o}{R_2}$$

$$e_o = -\frac{R_2}{R_1} e_i \dots \dots \dots (4.2)$$

The output is negative. Thus the circuit shown is an inverting amplifier with gain $\left(\frac{R_2}{R_1}\right)$. If $R_1 = R_2$, then the op amp circuit shown acts as an inverter.

Noninverting amplifier

Look at the circuit given in figure 4.9. Here the negative terminal is connected to the ground. The input voltage e_i is given to the positive terminal with respect to the ground.

$$i_1 = \frac{0 - e'}{R_1}, \quad i_2 = \frac{e' - e_o}{R_2}$$

$$i_1 = i_2 \gggg \gg \frac{0 - e'}{R_1} = \frac{e' - e_o}{R_2}$$

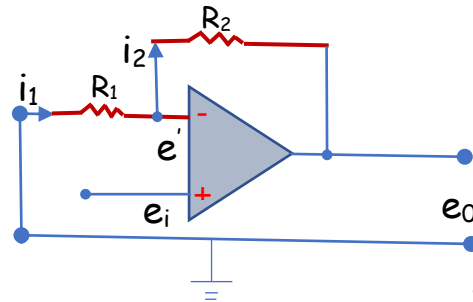


Fig. 4.9

$$-R_2 e' = R_1 (e' - e_o) \gggg \gg (R_1 + R_2) e' = R_1 e_o$$

$$e' = \frac{R_1}{(R_1 + R_2)} e_o$$

Referring to fig.4.7

$$e_o = K(e_2 - e_1) = K(e_i - e') = K \left(e_i - \frac{R_1}{R_1 + R_2} e_o \right)$$

$$e_o + K \frac{R_1}{(R_1 + R_2)} e_o = K e_i$$

$$e_o \left(\frac{1}{K} + \frac{R_1}{R_1 + R_2} \right) = e_i$$

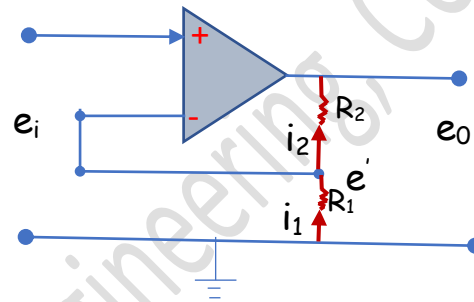


Fig. 4.10

Since K is very large, $\frac{1}{K}$ can be neglected. Hence,

$$e_o = \left(\frac{R_1 + R_2}{R_1} \right) e_i = \left(1 + \frac{R_2}{R_1} \right) e_i \dots \dots \dots (4.3)$$

Now the circuit acts as a noninverting amplifier with gain $\left(1 + \frac{R_2}{R_1} \right)$; The circuit can be also drawn as shown in figure 4.10.

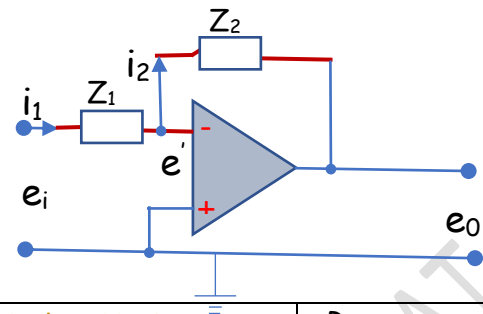
Similar to the case of electrical circuits discussed above, we can also apply the impedance approach to the op amp circuits to obtain their transfer functions

Consider the figure shown. Z_1 and Z_2 are the impedances connected. We can write

$$i_1 = \frac{e_i - e'}{Z_1}, \quad i_2 = \frac{e' - e_o}{Z_2}$$

Hence we can write $e_o = -\frac{Z_2}{Z_1} e_i$

$$E_o(s) = -\frac{Z_2(s)}{Z_1(s)} E_i(s) \dots \dots \dots (4.4)$$



If we have only a resistor R connected across a voltage V , then $V = iR$

Taking Laplace Transform, $V(s) = I(s) R = I(s) Z(s)$ where $Z(s) = R \dots \dots \dots (4.5a)$

If we have only a capacitor C connected across a voltage, $V = \frac{1}{C} \int i dt$

Taking Laplace transform, $V(s) = \frac{1}{C} \frac{I(s)}{s} = I(s) Z(s)$ where $Z(s) = \frac{1}{Cs} \dots \dots \dots (4.5b)$

If we have a resistor and a capacitor in series $V = V_R + V_C = iR + \frac{1}{C} \int i dt$

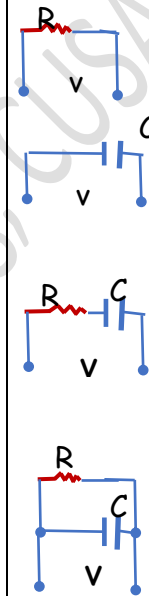
Taking Laplace transform, $V_o(s) = RI(s) + \frac{1}{C} \frac{I(s)}{s} = \frac{RCs+1}{Cs} I(s) = I(s) Z(s)$

Where $Z(s) = \frac{1+RCs}{Cs} \dots \dots \dots (4.5c)$

If we have a resistor and a capacitor in parallel across a voltage V , then we can write $I = I_R + I_C = \frac{V}{R} + C \frac{dv}{dt}$

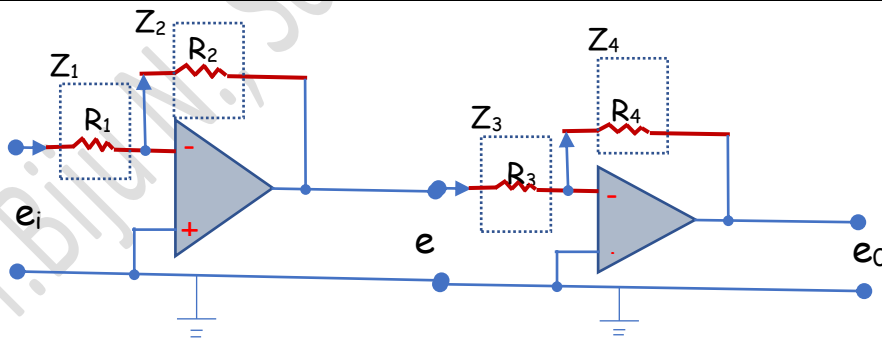
Taking Laplace Transform, $I(s) = \frac{V(s)}{R} + Cs V(s) = \left(\frac{1+RCs}{R} \right) V(s)$

$V(s) = \left(\frac{R}{1+RCs} \right) I(s) = I(s) Z(s)$ where $Z(s) = \frac{R}{1+RCs} \dots \dots \dots (4.5d)$



Proportional Controller

Consider the op amp circuit shown in fig. 4.11 where two op amps are connected in series. Let e be the output voltage of op amp 1, e_i the input voltage, and e_o the output voltage.



From equation (4.2), we can $e = -\frac{R_2}{R_1} e_i$, and $e_o = -\frac{R_4}{R_3} e$

$$e_o = -\left(\frac{R_4}{R_3} \right) \left(-\frac{R_2}{R_1} \right) e_i = \frac{R_4 R_2}{R_3 R_1} e_i$$

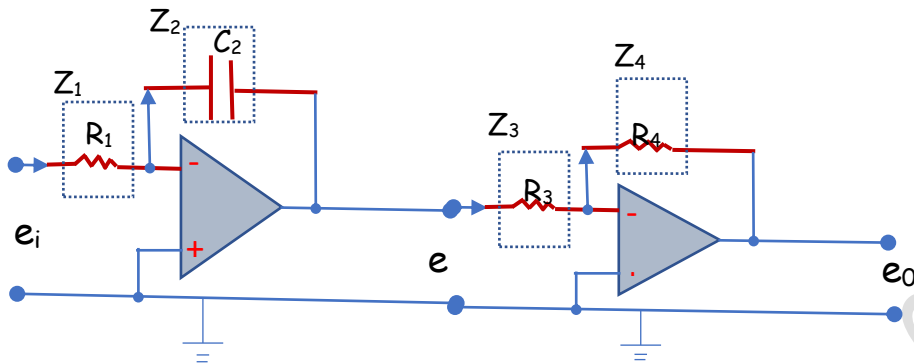
Taking Laplace transform $\frac{E_o(s)}{E_i(s)} = \frac{R_4 R_2}{R_3 R_1} = K_p$, the adjustable constant

We get a proportional controller.

If we use the impedance approach, $\frac{E_o(s)}{E_i(s)} = \frac{Z_4(s) Z_2(s)}{Z_3(s) Z_1(s)} = \frac{R_4 R_2}{R_3 R_1} = K_p$ where $Z_1(s) = R_1$, $Z_2(s) = R_2$, $Z_3(s) = R_3$, $Z_4(s) = R_4$. (Refer eq. 4.5a-d)

Integral Controller

Consider the op amp circuit shown. The resistance R_2 is replaced by a capacitance C_2 . Let us consider the impedance approach. $Z_1(s) = R_1$, $Z_2(s) = \frac{1}{C_2 s}$, $Z_3(s) = R_3$, $Z_4(s) = R_4$. (Refer eq. 4.5a-d)



Therefore from equation 4.4,

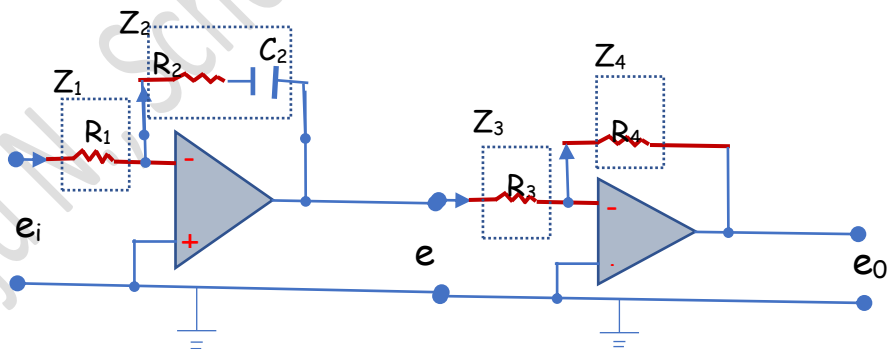
$$E(s) = -\frac{Z_2(s)}{Z_1(s)} E_i(s), \quad E_o(s) = -\frac{Z_4(s)}{Z_3(s)} E(s)$$

$$E_o(s) = -\frac{Z_4(s)}{Z_3(s)} \left(-\frac{Z_2(s)}{Z_1(s)}\right) E_i(s) = \frac{R_4}{R_3} \frac{1}{R_1} \frac{1}{C_2 s} E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \frac{R_4}{R_3} \frac{1}{R_1} \frac{1}{C_2 s} = \left(\frac{R_4}{R_3 R_1 C_2} \right) \frac{1}{s} = \frac{K_i}{s}$$

Proportional-plus-Integral Controller

Consider the op amp circuit shown. The resistance R_2 is in series with a capacitance C_2 . Let us consider the impedance approach. $Z_1(s) = R_1$, $Z_2(s) = \frac{1 + R_2 C_2 s}{C_2 s}$, $Z_3(s) = R_3$, $Z_4(s) = R_4$. (Refer Eq. 4.5a-d)



Therefore from equation 4.4,

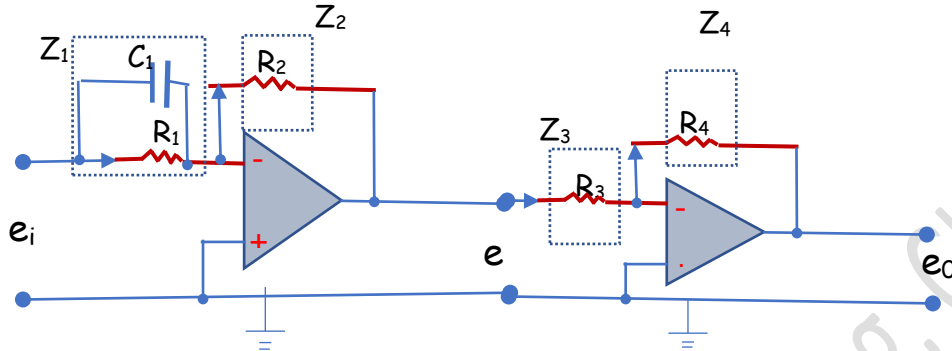
$$E(s) = -\frac{Z_2(s)}{Z_1(s)} E_i(s), \quad E_o(s) = -\frac{Z_4(s)}{Z_3(s)} E(s)$$

$$E_o(s) = -\frac{Z_4(s)}{Z_3(s)} \left(-\frac{Z_2(s)}{Z_1(s)}\right) E_i(s) = \left(\frac{R_4}{R_3}\right) \left(\frac{1 + R_2 C_2 s}{R_1 C_2 s}\right) E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \left(\frac{R_4}{R_3}\right) \left(\frac{1 + R_2 C_2 s}{R_1 C_2 s}\right) = \left(\frac{R_2 R_4}{R_1 R_3} + \frac{R_4}{R_3 R_1 C_2} \frac{1}{s}\right) = K_P + \frac{K_i}{s}$$

Proportional-plus-Derivative controller

Consider the op amp circuit shown. The resistance R_1 is in series with a capacitance C_1 . Let us consider the impedance approach. $Z_1(s) = \frac{R_1}{1+R_1C_1s}$, $Z_2(s) = R_2$, $Z_3(s) = R_3$, $Z_4(s) = R_4$. (Refer Eq. 4.5a-d)



Therefore from equation 4.4,

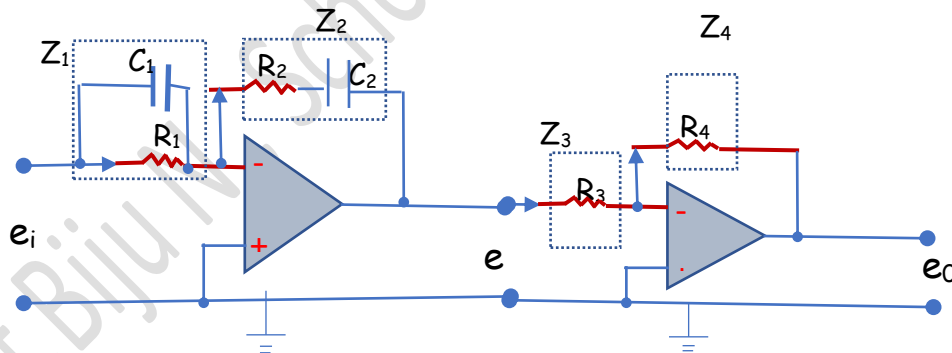
$$E(s) = -\frac{Z_2(s)}{Z_1(s)} E_i(s), \quad E_o(s) = -\frac{Z_4(s)}{Z_3(s)} E(s)$$

$$E_o(s) = -\frac{Z_4(s)}{Z_3(s)} \left(-\frac{Z_2(s)}{Z_1(s)} \right) E_i(s) = \left(\frac{R_4}{R_3} \right) \left(\frac{R_2}{\frac{R_1}{1+R_1C_1s}} \right) E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \left(\frac{R_4}{R_3} \right) R_2 \left(\frac{1+R_1C_1s}{R_1} \right) = \left(\frac{R_2R_4}{R_1R_3} + \frac{R_2R_4C_1}{R_3} s \right) = K_P + K_d s$$

Proportional-plus-Integral-plus-Derivative controller

$Z_1(s) = \frac{R_1}{1+R_1C_1s}$, $Z_2(s) = \frac{1+R_2C_2s}{C_2s}$, $Z_3(s) = R_3$, $Z_4(s) = R_4$. (Refer Eq. 4.5a-d)



Therefore from equation 4.4,

$$E(s) = -\frac{Z_2(s)}{Z_1(s)} E_i(s), \quad E_o(s) = -\frac{Z_4(s)}{Z_3(s)} E(s)$$

$$E_o(s) = -\frac{Z_4(s)}{Z_3(s)} \left(-\frac{Z_2(s)}{Z_1(s)} \right) E_i(s) = \left(\frac{R_4}{R_3} \right) \left(\frac{\frac{1+R_2C_2s}{C_2s}}{\frac{R_1}{1+R_1C_1s}} \right) E_i(s)$$

$$\frac{E_o(s)}{E_i(s)} = \left(\frac{R_4}{R_3} \right) \left(\frac{1}{C_2s} + R_2 \right) \left(\frac{1}{R_1} + C_1s \right) = \left(\frac{R_4}{R_3} \right) \left(\frac{1}{R_1C_2s} + \frac{R_2}{R_1} + \frac{C_1}{C_2} + R_2C_1s \right)$$

$$= \left(\frac{R_4}{R_3} \right) \left(\frac{R_2}{R_1} + \frac{C_1}{C_2} \right) + \frac{\frac{R_4}{R_1C_2R_3}}{s} + \frac{R_4R_2C_1}{R_3} s = K_P + \frac{K_i}{s} + K_d s$$

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